RTL TO GDS-II WORKSHOP

Project Report on

"RTL to GDS-II Implementation"

Submitted by

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Problem Statement:

1. Implement the equation Y = floor(log2(X)) where X is an 8-bit unsigned number
2. Conditions: - a) Do not use OR cell.
   b) Use L shape scenario (No. 3) for floor planning.

3. Generate GDS-II layout of design using Synopsys-flowscripts and report the following:

   a) Design area
   b) Clock frequency (for a positive setup slack value but less than 1)
   c) Power consumption

RTL and Test-bench – Verilog

1. RTL Source code:

File name- log2_seq.v

```
`timescale 1ns / 1ps

module log2_seq (
    input clk,
    input [7:0] X,
    output reg [2:0] Y
);
reg [2:0] Y_comb;

always @(*) begin
    if (X[7])       Y_comb = 3'd7;
    else if (X[6])  Y_comb = 3'd6;
    else if (X[5])  Y_comb = 3'd5;
    else if (X[4])  Y_comb = 3'd4;
    else if (X[3])  Y_comb = 3'd3;
    else if (X[2])  Y_comb = 3'd2;
    else if (X[1])  Y_comb = 3'd1;
    else if (X[0])  Y_comb = 3'd0;
    else            Y_comb = 3'd0;
end

always @(posedge clk) begin
    Y <= Y_comb;
end
endmodule
```

2. Test-Bench:

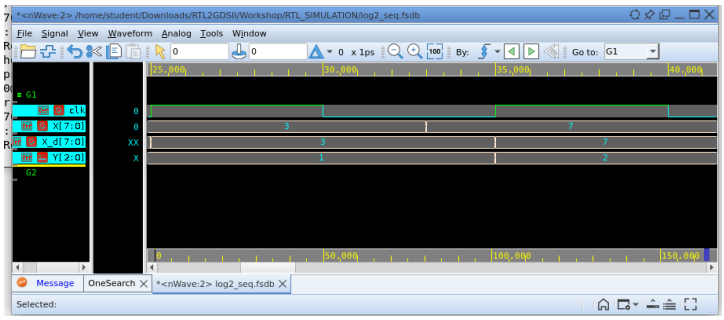
File name: log2_seq_tb.v

```
`timescale 1ns / 1ps

module log2_seq_tb;

reg clk;
reg [7:0] X;
reg [7:0] X_d;
wire [2:0] Y;

log2_seq uut (
    .clk(clk),
    .X(X),
    .Y(Y)
);

always #5 clk = ~clk;

always @(posedge clk) begin
    X_d <= X;
end

initial begin
    clk = 0;
    X = 0;

    #3 X = 8'd1;
    #10 X = 8'd2;
```

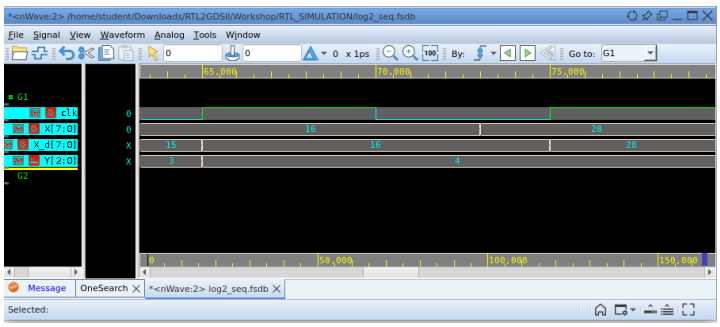
```
    #10 X = 8'd3;
    #10 X = 8'd7;
    #10 X = 8'd8;
    #10 X = 8'd15;
    #10 X = 8'd16;
    #10 X = 8'd28;
    #10 X = 8'd37;
    #10 X = 8'd99;
    #10 X = 8'd127;
    #10 X = 8'd128;
    #10 X = 8'd200;
    #10 X = 8'd255;
    #10 X = 8'd0;

    #20 $finish;
end

always @(posedge clk) begin
    $display("Time=%0t  X_d=%0d  Y=%0d", $time, X_d, Y);
end

endmodule
```

3. Timing Diagram:

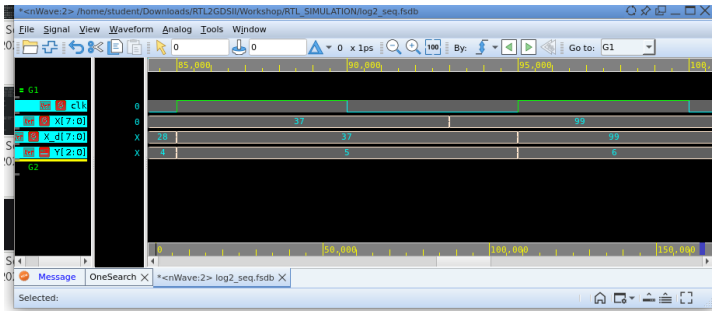
- When X=3, Y=1
- When X=7, Y=2



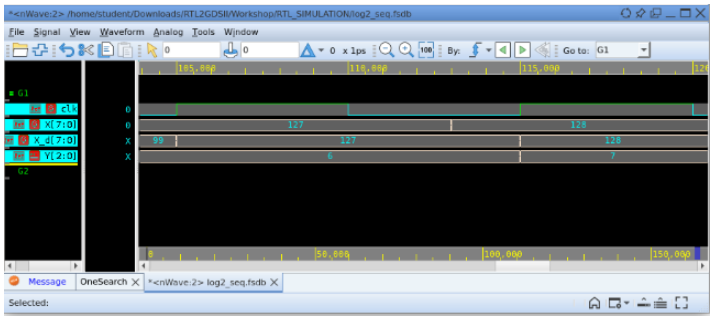
- When X=16, Y=4
- When X=28, Y=4



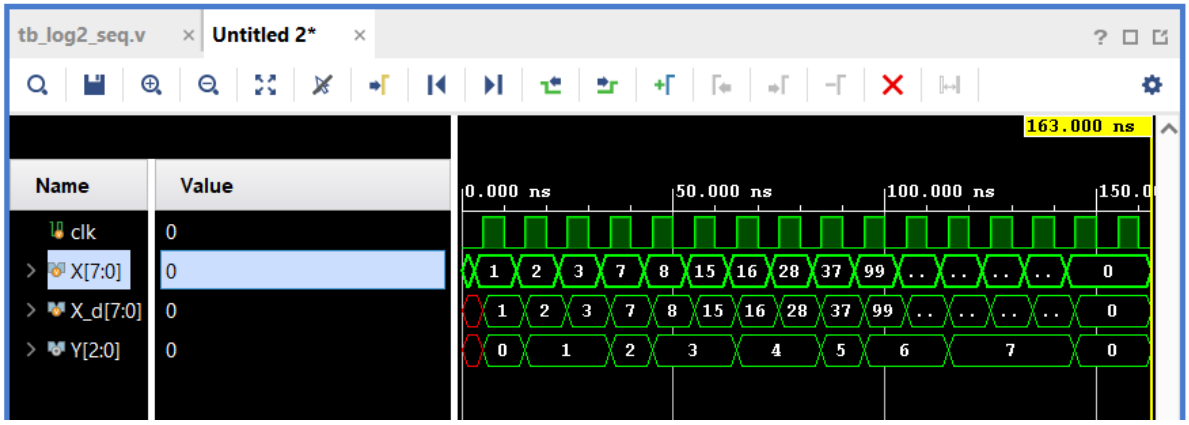
- When X=37, Y=5
- When X=99, Y=6



- When X=127, Y=6
- When X=128, Y=7



4. Timing Diagram(On Vivado):



Logic Synthesis

1. Changes in common_setup.tcl File:

a) DESIGN_NAME : log2_seq

2. Constraint File:

File name: log2_seq.sdc

```
# Clock definition
 create_clock -period 10 -name "clk" [get_ports clk]
 set_clock_uncertainty 0.2 [get_clocks clk]
 set_clock_transition  0.1 [get_clocks clk]

# Input/Output delays
 set_input_delay  2.0 -clock clk [all_inputs]
 set_output_delay 2.0 -clock clk [all_outputs]

# False paths
# set_false_path -from [get_ports rst_n]
```

3. Design Compiler TCL commands:

Filename: dc_script.tcl

```
#==================================================================
# Synopsys Design Compiler
#==================================================================

#-----------------------------------------------------------
# 1. SETUP & ENVIRONMENT
```

```
#-------------------------------------------------------------

set DESIGN_NAME      "log2_seq"
set RTL_DIR          "../RTL/"
set CONSTRAINTS_FILE "../constraints/${DESIGN_NAME}.sdc"
set LIBRARY_DIR      "../ref/lib/stdcell_rvt/"
set OUTPUT_DIR       "./outputs"
set REPORTS_DIR      "./reports"

# Target library (technology .db files)
set target_library   "$LIBRARY_DIR/saed32rvt_ss0p7vn40c.db"
set link_library     "* $LIBRARY_DIR/saed32rvt_ss0p7vn40c.db
$LIBRARY_DIR/saed32rvt_ff1p16v125c.db"

# Create output directories
file mkdir $OUTPUT_DIR
file mkdir $REPORTS_DIR

#-------------------------------------------------------------
# 2. READ RTL SOURCE FILES
#-------------------------------------------------------------

# Option A: Read Verilog
read_verilog [list \
    $RTL_DIR/${DESIGN_NAME}_rtl.v \
]

# Option B: Read SystemVerilog
# read_file -format sverilog [list \
```

```
#     $RTL_DIR/top_module.sv \
#     $RTL_DIR/submodule_a.sv \
# ]

# Option C: Read VHDL
# read_vhdl [list \
#     $RTL_DIR/top_module.vhd \
# ]

# Set the top-level design
current_design $DESIGN_NAME

#-------------------------------------------------------------
# 3. LINK THE DESIGN
#-------------------------------------------------------------

link

#-------------------------------------------------------------
# 4. SET DON'T USE CELLS
#-------------------------------------------------------------

#set_dont_use [get_lib_cells */FADD*]
#set_dont_use [get_lib_cells */HADD*]
#set_dont_use [get_lib_cells */AO*]
#set_dont_use [get_lib_cells */OA*]
#set_dont_use [get_lib_cells */NAND*]
#set_dont_use [get_lib_cells */XOR*]
#set_dont_use [get_lib_cells */NOR*]
```

```
#set_dont_use [get_lib_cells */XNOR*]
#set_dont_use [get_lib_cells */MUX*]
set_dont_use [get_lib_cells */OR*]

#-------------------------------------------------------------
# 5. APPLY CONSTRAINTS (SDC)
#-------------------------------------------------------------

source $CONSTRAINTS_FILE

#-------------------------------------------------------------
# 6. COMPILE (SYNTHESIS)
#-------------------------------------------------------------

# Standard compile
# compile -map_effort medium -area_effort medium

# Ultra compile
  compile_ultra

# Incremental compile for further optimization (optional)
# compile_ultra -incremental

# Fix hold violations
#set_fix_hold [all_clocks]
#compile_ultra -incremental -only_hold_time

#-------------------------------------------------------------
# 7. POST-COMPILE OPTIMIZATIONS
```

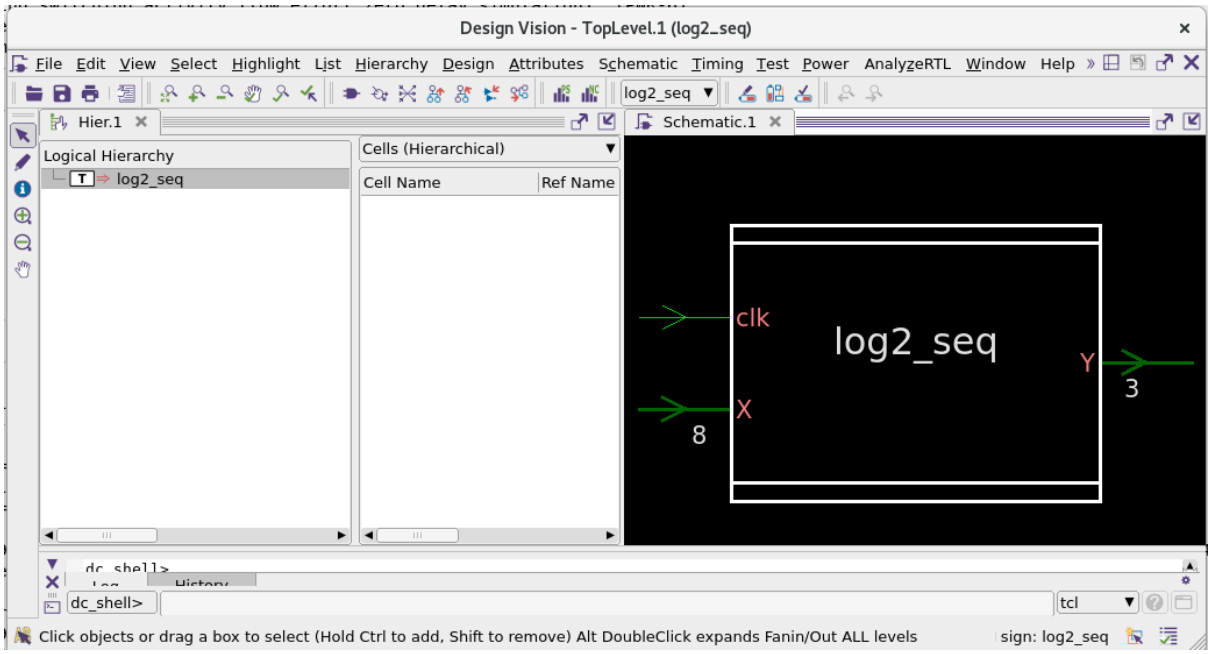
```
#-------------------------------------------------------------

# Remove unconnected ports/nets
remove_unconnected_ports [get_cells -hierarchical *]

#-------------------------------------------------------------
# 8. REPORTS
#-------------------------------------------------------------

# Timing reports
report_timing -path full -delay max -max_paths 10 \
   > $REPORTS_DIR/timing_setup.rpt

report_timing -path full -delay min -max_paths 10 \
   > $REPORTS_DIR/timing_hold.rpt

# Area report
report_area -hierarchy \
   > $REPORTS_DIR/area.rpt

# Power report
report_power -hierarchy \
   > $REPORTS_DIR/power.rpt

# Design rule checks
report_constraint -all_violators \
   > $REPORTS_DIR/constraints.rpt

report_cell \
```

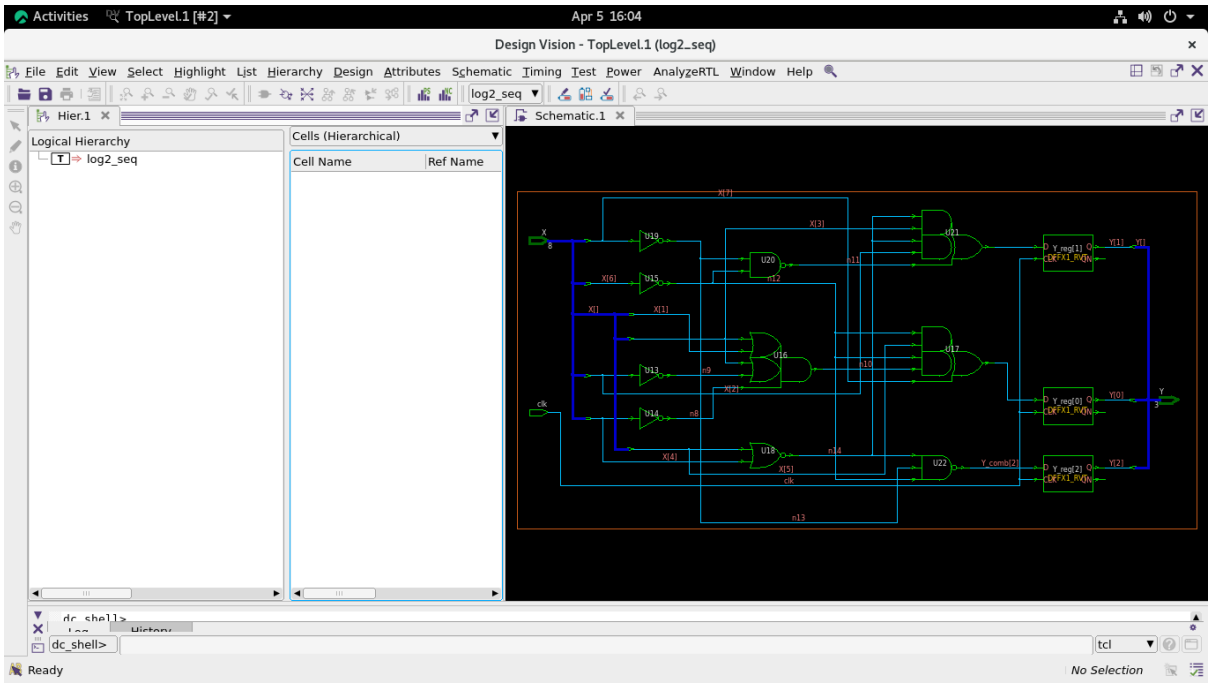
```
    > $REPORTS_DIR/cells.rpt

report_net \
    > $REPORTS_DIR/nets.rpt

# QoR summary
report_qor \
    > $REPORTS_DIR/qor.rpt

#-------------------------------------------------------------
# 9. WRITE OUTPUTS
#-------------------------------------------------------------

# Remove multi-port nets and constant logic assign statements from netlist
#set_fix_multiple_port_nets -all -buffer_constants

# Write gate-level netlist (Verilog)
write -format verilog -hierarchy \
    -output $OUTPUT_DIR/${DESIGN_NAME}_netlist.v

# Write SDC constraints
write_sdc $OUTPUT_DIR/${DESIGN_NAME}.sdc

#-------------------------------------------------------------
# 10. DONE
#-------------------------------------------------------------

echo "============================================"
echo " Synthesis COMPLETE: $DESIGN_NAME"
```

echo "==="

4. Generated outputs:

Schematics:





Reports Summary:

Parameter	Value
Power Consumption	1.11 µW
Cell Area	39.90 μm^2
Design Area	43.11 μm^2
Total Nets	21
Total Cells	13

```
==================================================================================================
| Flow Information                                                                               |
--------------------------------------------------------------------------------------------------
| Flow         | Design Compiler WLM                                                             |
| Command Line | compile_ultra                                                                   |
==================================================================================================
| Design Information                                                  | Value                    |
==================================================================================================
| Number of Scenarios                                                 | 0                        |
| Leaf Cell Count                                                     | 33                       |
| Number of User Hierarchies                                          | 0                        |
| Sequential Cell Count                                               | 3                        |
| Macro Count                                                         | 0                        |
| Number of Power Domains                                             | 0                        |
| Number of Path Groups                                               | 2                        |
| Number of VT Class                                                  | 1                        |
| Number of Clocks                                                    | 1                        |
| Number of Dont Touch Cells                                          | 29                       |
| Number of Dont Touch Nets                                           | 0                        |
| Number of Size Only Cells                                           | 0                        |
| Design with UPF Data                                                | false                    |
==================================================================================================
```

Physical Design(ICC2 Shell)

1. Setup

```
set DESIGN_NAME   "log2_seq"

# Input files (from DC output)
set NETLIST_FILE  "../DC/outputs/${DESIGN_NAME}_netlist.v"
set SDC_FILE      "../DC/outputs/${DESIGN_NAME}.sdc"

# PDK / library paths
set PDK_PATH      "../ref"
set LIB_DIR       "$PDK_PATH/lib/stdcell_rvt"
set TECH_DIR      "$PDK_PATH/tech"

# Output directories
set OUTPUT_DIR    "./outputs"
set REPORTS_DIR   "./reports"

# Create output directories
file mkdir $OUTPUT_DIR
file mkdir $REPORTS_DIR

puts "INFO: Setup complete for design: $DESIGN_NAME"
```

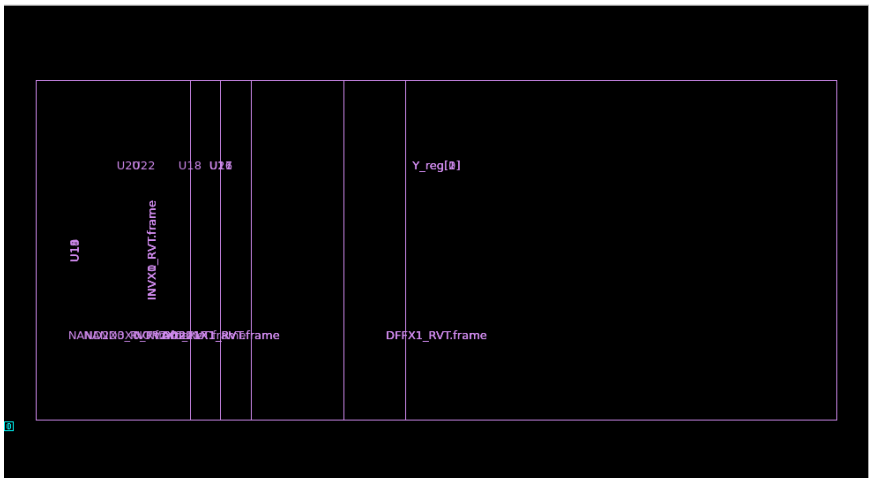
2. Netlist Read

```
source scripts/01_setup.tcl

# 2a. Create the in-memory library and attach the NDM reference

create_lib \
    -ref_lib $PDK_PATH/lib/ndm/saed32rvt_c.ndm \
    log2_seq_LIB

# 2b. Read the synthesised gate-level netlist

read_verilog $NETLIST_FILE \
    -library log2_seq_LIB \
    -design  $DESIGN_NAME  \
    -top     $DESIGN_NAME

# 2c. Link and constrain the design

link_block
read_sdc $SDC_FILE

# 2d. Pre-placement design check

check_design -checks pre_placement_stage \
    > $REPORTS_DIR/check_design_pre_place.rpt

save_block
save_lib
```

puts "INFO: Netlist read and design linked successfully."

start_gui

Design Layout



3. Floor planning

#scenario3:

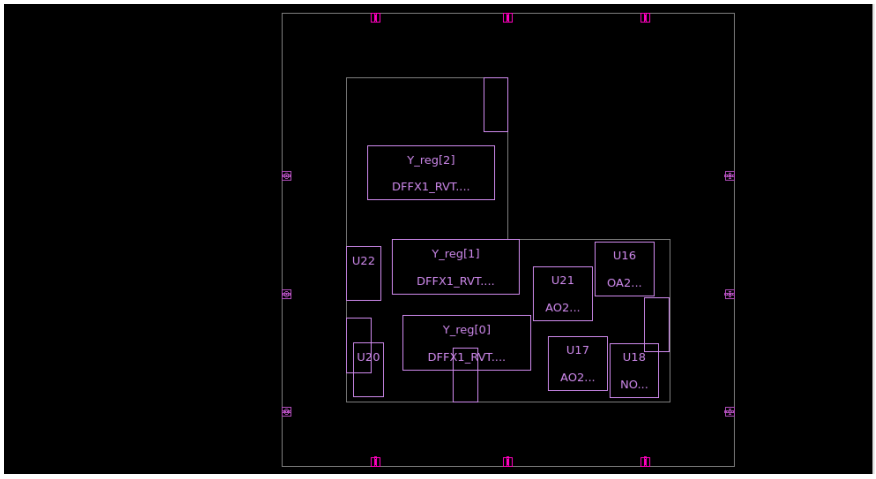
initialize_floorplan -shape L -core_utilization 0.5 -core_offset 2 -coincident_boundary false

set_individual_pin_constraints -ports [get_ports X] -sides 6

place_pins -self

create_placement -floorplan -effort medium

Design Layout



4. Power Planning

source scripts/01_setup.tcl

4a. Create and connect power / ground nets

create_net -power {VDD}

create_net -ground {VSS}

connect_pg_net -all_blocks -automatic

4b. Power / ground rings

create_pg_ring_pattern core_ring_pattern -horizontal_layer M7 -horizontal_width .4 -horizontal_spacing .3 -vertical_layer M8 -vertical_width .4 -vertical_spacing .3

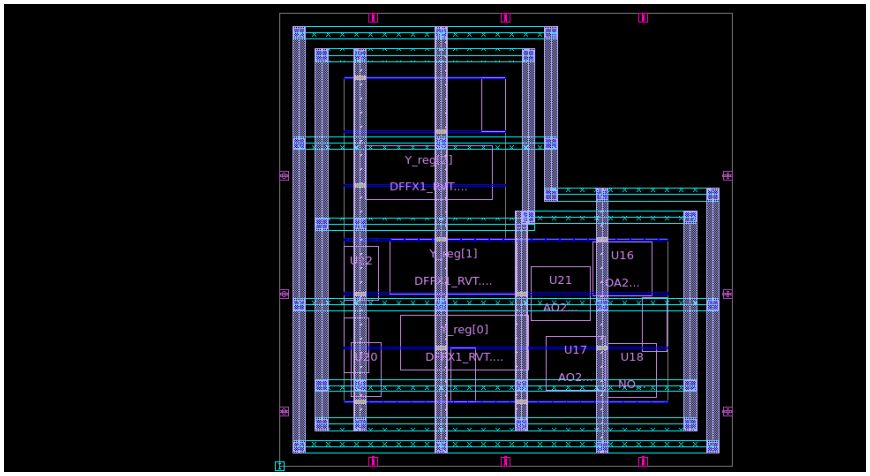
set_pg_strategy core_power_ring -core -pattern {{name : core_ring_pattern}{nets : {VDD VSS}}{offset : {.5 .5}}}

compile_pg -strategies core_power_ring

4c. Power mesh

```
create_pg_mesh_pattern mesh -layers { {{vertical_layer: M6}{width: .34} {spacing: interleaving}{pitch: 5} {offset: .5}} {{horizontal_layer: M7}{width: .38} {spacing: interleaving} {pitch: 5} {offset: .5}} {{vertical_layer: M8}{width: .38} {spacing: interleaving}{pitch: 5} {offset: .5}}}

set_pg_strategy core_mesh -pattern { {pattern:mesh} {nets: VDD VSS}} -core -extension {stop: innermost_ring}

compile_pg -strategies core_mesh

# 4d. Standard-cell power rails

create_pg_std_cell_conn_pattern std_cell_rail -layers {M1} -rail_width 0.06

set_pg_strategy rail_strat -core -pattern {{name: std_cell_rail} {nets: VDD VSS} }

compile_pg -strategies rail_strat

# 4e. Connectivity check

check_pg_connectivity \
   > $REPORTS_DIR/pg_connectivity.rpt

save_block
save_lib

puts "INFO: Power planning complete."
```

Design layout:



5. Placement

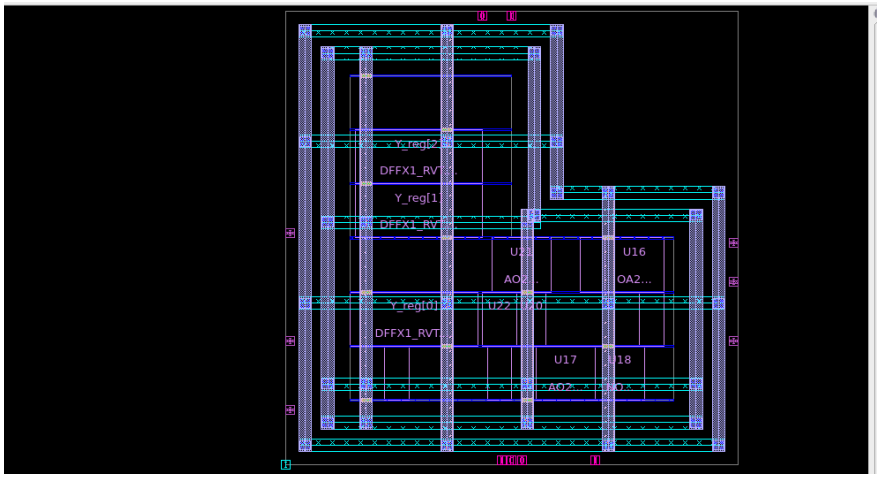
```
source scripts/01_setup.tcl

# 5a. Timing mode / corner / scenario setup

set mode1     "func"
set corner1   "nom"
set scenario1 "${mode1}::${corner1}"

remove_modes -all
remove_corners -all
remove_scenarios -all

create_mode   $mode1
create_corner $corner1
create_scenario -name func::nom -mode func -corner nom

current_mode     func
current_scenario func::nom
```

```
# Load SDC
source $SDC_FILE

# 5b. Parasitic technology

set parasitic1 "p1"
set tluplus_file$parasitic1 "$PDK_PATH/tech/star_rcxt/saed32nm_1p9m_Cmax.tluplus"
set layer_map_file$parasitic1 "$PDK_PATH/tech/star_rcxt/saed32nm_tf_itf_tluplus.map"

set parasitic2 "p2"
set tluplus_file$parasitic2 "$PDK_PATH/tech/star_rcxt/saed32nm_1p9m_Cmin.tluplus"
set layer_map_file$parasitic2 "$PDK_PATH/tech/star_rcxt/saed32nm_tf_itf_tluplus.map"

read_parasitic_tech -tlup $tluplus_filep1 -layermap $layer_map_filep1 -name p1
read_parasitic_tech -tlup $tluplus_filep2 -layermap $layer_map_filep2 -name p2

set_parasitic_parameters -late_spec $parasitic1 -early_spec $parasitic2

# 5c. Placement app options

set_app_options -name place.coarse.continue_on_missing_scandef -value true

# 5d. Run placement

place_pins -self
place_opt
legalize_placement
```

5e. Reports

```
check_legality -verbose \
    > $REPORTS_DIR/check_legality.rpt

report_timing -delay max -max_paths 10 \
    > $REPORTS_DIR/timing_post_place.rpt

report_qor -summary \
    > $REPORTS_DIR/qor_post_place.rpt

save_block -as log2_seq_placement
save_lib

puts "INFO: Placement complete."
```

Design layout:

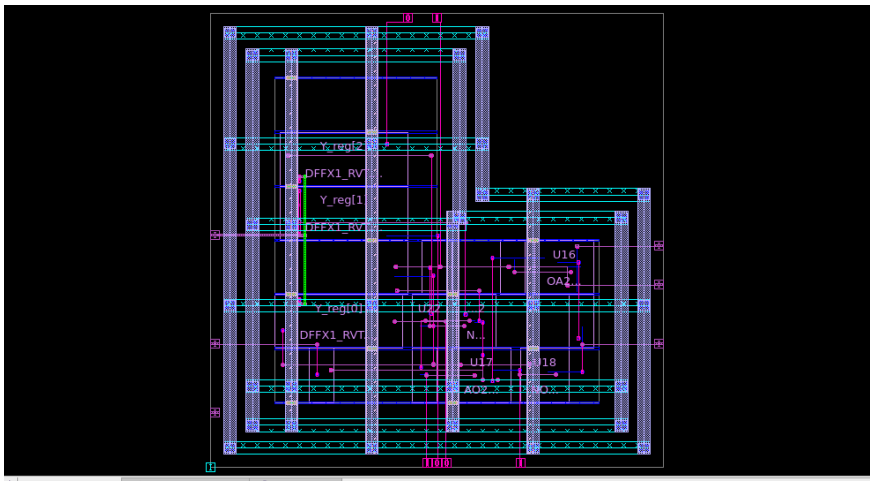


6. Clock Tree synthesis

source scripts/01_setup.tcl

```
check_design -checks pre_clock_tree_stage

# 6a. CLOCK TREE SYNTHESIS (CTS)

synthesize_clock_tree
set_app_options -name cts.optimize.enable_local_skew -value true
set_app_options -name cts.compile.enable_local_skew -value true
set_app_options -name cts.compile.enable_global_route -value false
set_app_options -name clock_opt.flow.enable_ccd -value true

clock_opt -to build_clock
clock_opt -from route_clock -to route_clock
clock_opt

# 6e. Reports

report_clock_qor > $REPORTS_DIR/clock_qor.rpt

report_clock_settings > $REPORTS_DIR/clock_settings.rpt

report_timing -delay max -max_paths 10 > $REPORTS_DIR/timing_post_cts.rpt

report_timing -delay min -max_paths 10 > $REPORTS_DIR/timing_hold_post_cts.rpt

save_block -as log2_seq_cts
save_lib

puts "INFO: Clock tree synthesis complete."
```

Design layout:



7. Routing:

source scripts/01_setup.tcl

7a. Global route options

set_app_options -name route.global.timing_driven -value true

set_app_options -name route.global.crosstalk_driven -value false

7b. Track assignment options

set_app_options -name route.track.timing_driven -value true

set_app_options -name route.track.crosstalk_driven -value true

7c. Detail route options

set_app_options -name route.detail.timing_driven -value true

#set_app_options -name route.detail.save_after_iterations -value false

set_app_options -name route.detail.force_max_number_iterations -value false

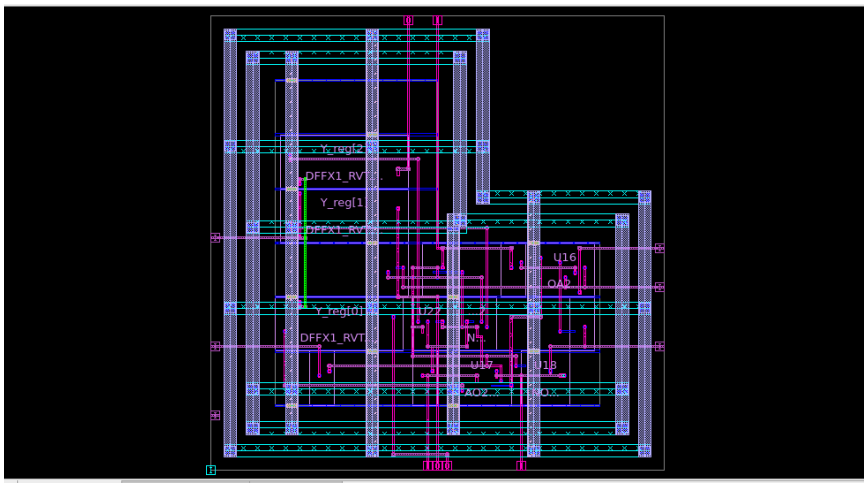
set_app_options -name route.detail.antenna -value true

set_app_options -name route.detail.antenna_fixing_preference -value use_diodes

```
set_app_options -name route.detail.diode_libcell_names -value */ANTENNA_RVT

# 7d. Atomic routing sequence

route_global

route_track

route_detail

# 7e. Post-route optimisation

route_opt

# 7f. Routing checks and reports

check_routes \
    > $REPORTS_DIR/check_routes.rpt

report_timing -delay max -max_paths 10 \
    > $REPORTS_DIR/timing_post_route.rpt

report_timing -delay min -max_paths 10 \
    > $REPORTS_DIR/timing_hold_post_route.rpt

# 7g. Write routed outputs

write_verilog $OUTPUT_DIR/${DESIGN_NAME}.routed.v
write_sdc -output $OUTPUT_DIR/${DESIGN_NAME}.routed.sdc
```

```
write_parasitics \
    -format spef \
    -output $OUTPUT_DIR/${DESIGN_NAME}.spef

save_block
save_lib

puts "INFO: Routing complete."
```

Design layout:



8. Report Files:

```
Area
----------------------------------------
Combinational Area:                 20.59
Noncombinational Area:              19.82
Buf/Inv Area:                        5.08
Total Buffer Area:                   0.00
Total Inverter Area:                 5.08
Macro/Black Box Area:                0.00
Net Area:                               0
Net XLength:                        65.42
Net YLength:                        69.67
----------------------------------------
Cell Area (netlist):                              40.41
Cell Area (netlist and physical only):            40.41
Net Length:                        135.08
```

Powerplanning and Placement

Stack	7.72 ns
Power consumption	1.11 uW
Area	43.11 um^2
Connectivity	Fully Connected

Clock Tree Synthesis:

Stack	7.72 ns
Power consumption	1.11 µW
Area	43.11 um^2

Routing

Stack	7.72 ns
Power consumption	1.11 µW
Area	43.11 um^2
Connectivity	Fully Connected
Routing Verification	No Open Net

Static Timing Analysis

1. Late corners:

```
set report_default_significant_digits 6
set link_path "../ref/lib/stdcell_rvt/saed32rvt_ss0p7vn40c.db"

set DESIGN_NAME "log2_seq"
set REPORTS_DIR "./reports"
file mkdir $REPORTS_DIR

read_verilog "../ICCII/outputs/${DESIGN_NAME}.routed.v"
link_design
current_design $DESIGN_NAME

read_sdc "../ICCII/outputs/${DESIGN_NAME}_final.sdc"

read_parasitics "../ICCII/outputs/${DESIGN_NAME}_func::nom.spef.p1_125.spef"

update_timing -full
```

```
report_timing
report_design

report_timing -path full -delay max -max_paths 10 \
   > $REPORTS_DIR/timing_setup.rpt

report_timing -path full -delay min -max_paths 10 \
   > $REPORTS_DIR/timing_hold.rpt
```

```
****************************************
Report : timing
        -path_type full
        -delay_type min
        -slack_lesser_than 0.000000
        -max_paths 10
        -sort_by slack
Design : log2_seq
Version: W-2024.09
Date   : Sun Apr  5 15:51:35 2026
****************************************

No paths with slack less than 0.000000.
```

2. Early corners:

```
set report_default_significant_digits 6 ;
set link_path "../ref/lib/stdcell_rvt/saed32rvt_ff1p16v125c.db"

set DESIGN_NAME "full_adder"
set REPORTS_DIR "./reports"
file mkdir $REPORTS_DIR

read_verilog "../ICCII/outputs/${DESIGN_NAME}.routed.v"
```

```
link_design
current_design $DESIGN_NAME

read_sdc "../ICCII/outputs/${DESIGN_NAME}_final.sdc"

read_parasitics "../ICCII/outputs/${DESIGN_NAME}_func::nom.spef.p2_125.spef"

update_timing -full

report_timing
report_design

report_timing -path full -delay max -max_paths 10 \
   > $REPORTS_DIR/timing_setup.rpt

report_timing -path full -delay min -max_paths 10 \
   > $REPORTS_DIR/timing_hold.rpt
```

```
****************************************
Report : timing
        -path_type full
        -delay_type max
        -slack_lesser_than 0.000000
        -max_paths 10
        -sort_by slack
Design : prob1
Version: W-2024.09
Date   : Sun Apr  5 15:42:58 2026
****************************************
```

3. Report Content:

Stack	7.72 ns(Setup), 1.83 ns(Hold)
Violations	None
Timing Status	Met

Final Results

Cell Area	39.90 μm^2
Total Area	43.11 μm^2
Power Consumption	1.11µW
Clock Frequency	~100 MHz (10 ns clock)

Learnings

I. RTL Development:

- To implement a circuit using Behavioural modelling in Verilog HDL
- To simulate a Verilog testbench using VCS and Verdi tools

II. Logic Synthesis

- Using Design Compiler tool to build gate-level logic design of provided Verilog module • Setting up the constraints and updating constraints to adjust the outputs
- Understood the concepts like worst setup slack and process corners which create an impact on our design

III. Physical Design.

- Learned to create a floorplan and power plan for the physical design using IC Compiler tool, and to place cells based on the power plan
- Learned to check whether the design is connected or not to verify the above steps
- Learned to synthesize clock tree and checking its latency

• Learned routing process to connect each path physically in the design

IV. Static Timing Analysis

- Using Primetime tool to perform the Static Timing Analysis of the design
- Learned to interpret the reports generated by tools for analysis of design
- Learned to implement the VLSI Design Flow from RTL Development to STA with the help of Synopsys Tools